MCS-334	DATA WAREHOUSING & MINNING	L T P 3 1 0	4 Credits

Course Objectives:

CO1	To learn a basic concept of Data warehousing.
CO2	To learn various models of data warehousing
CO3	To understand the concept of data mining
CO4	To understand the concept of Data Compression. Statistical measures in large Databases
CO5	To understand basic OLAP functions

MODULE I

Foundation

Introduction to DATA Warehousing. Client/Server Computing model & Data Warehousing. Parallel processors & Cluster Systems. Distributed DBMS implementations. Client/Server RDBMS Solutions.

Data Warehousing

Data Warehousing Components. Building a Data Warehouse. Mapping the Data Warehousing to a Multiprocessor Architecture. DBMS Schemas for Decision Support. Data Extraction, cleanup & Transformation Tools. Metadata.

MODULE II

Business Analysis

Reporting & Query Tools & Applications. On line Analytical Processing (OLAP). Patterns & Models. Statistics. Artificial Intelligence.

MODULE III

Data Mining

Introduction to Data Mining. Decision Trees. Neural Networks. Nearest Neighbor & Clustering. Genetic Algorithms. Rule Induction. Selecting & Using the Right Technique.

Data visualization & Overall Perspective. Data Visualization. Putting it All Together. Appendices: A : Data Visualization. B : Big Data-Better Returns : Leveraging Your Hidden Data Assets to Improve ROI. C : Dr. E.F. Codd's 12 Guidelines for OLAP. D : Mistakes for Data warehousing Managers to Avoid.

References:

1. *Berson, "Data Warehousing, Data Mining & OLAP", Tata McGraw-Hill, New Delhi*
2. *Mallach, "Data Warehousing System", McGraw Hill.*
3. *Jiawei Han, Micheline Kamber "Data Mining: Concepts and Techniques", The Morgan Kaufmann Series in Data Management*
4. *Jim Gray, "Systems", Morgan Kaufmann Publishers, August 2000, ISBN 1-55860-489-8.*

Course Outcome:

CO1	Analyze the basic functions of data warehouse and data mining.
CO2	Design data warehouse with dimensional modelling and apply different operations.
CO3	Analyze OLAP functions
CO4	Analyze appropriate data mining algorithms to solve real world problems
CO5	Evaluate different data mining techniques like classification, prediction.